

Kea-Tiong Tang

professor

基本信息

姓名: Kea-Tiong Tang

所属院校: 台湾清华大学

地区: China-Taiwan

海外经历: 有

学科领域

人工智能

集成电路设计与集成系统

电子与计算机工程

生物医学工程

研究领域

深度学习与神经网络

低功耗电路与能量管理

低功耗设计与能耗优化

模拟与射频IC设计

生物信号处理与模式识别

混合信号集成技术

嵌入式系统与硬件设计

计算机体系结构

集成电路设计与制造

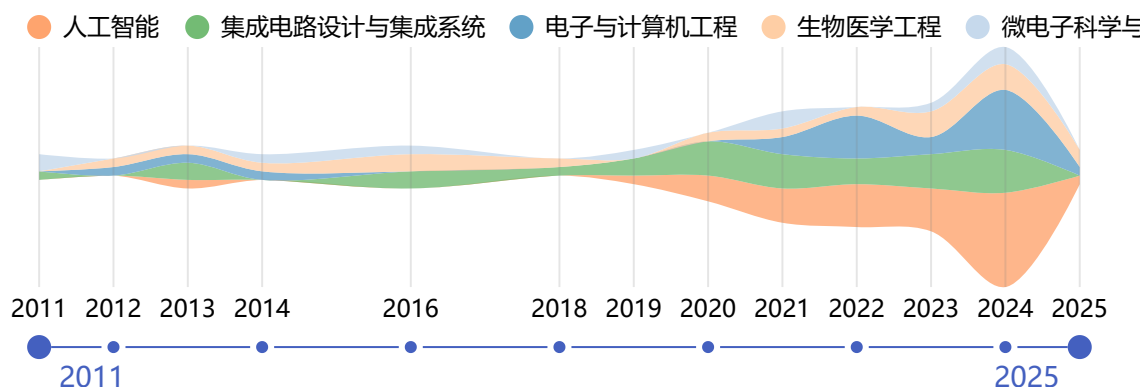
智能边缘计算

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个人简介

Kea-Tiong Tang is a professor in the Department of Electrical Engineering. He holds a Ph.D. from the California Institute of Technology. His research interests include Artificial Intelligence (AI) Chip, Neuromorphic chip, Biomedical signal processing, and more. His office tel. is (03)5162178. You can visit his personal website at <http://nbme.ee.nthu.edu.tw/>. The click number of his profile is 591.

近年发表的论文



Brain computer interfaces for cognitive enhancement in older people - challenges and applications: a systematic review

BMC GERIATRICS

Tsai, Ping-Chen; Akpan, Asangaedem; Tang, Kea-Tiong; Lakany, Heba

被引用次数: 0

2025

ANP-G: A 28-nm 1.04-pJ/SOP Sub-mm² Asynchronous Hybrid Neural Network Olfactory Processor Enabling Few-Shot Class-Incremental On-Chip Learning

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Huo, Dexuan; Zhang, Jilin; Dai, Xinyu; Zhang, Jian; Qian, Chunqi; Tang, Kea-Tiong; Chen, Hong

被引用次数: 0

2025

Creating a Sustainable Future Workshop: Cultivating a Green and Resilient Environment in CAS and Reliability for Security in Hardware

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Wang, Yi; Tang, Kea-Tiong

被引用次数: 0

2025

Exhaled Breath Analysis Using a Novel Electronic Nose for Different Respiratory Disease Entities

LUNG

Yu, Kai-Lun; Yang, Han-Ching; Lee, Chien-Feng; Wu, Shang-Yu; Ye, Zhong-Kai; Rai, Sujeet Kumar; Lee, Meng-Rui; Tang, Kea-Tiong; Wang, Jann-Yuan

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2025

Live Demonstration: Low Power and Real-Time Obstacle Avoidance System with CMOS Image Sensor and Spiking Neural Network

2024 IEEE THE 20TH ASIA PACIFIC CONFERENCE ON CIRCUITS AND SYSTEMS, APCCAS 2024

Huang, Wei-Chi; Yeh, Chen-Fu; Chen, Po-An; Tsai, Yu-Chih; Tang, Kea-Tiong

被引用次数: 0

2024

SUN: Dynamic Hybrid-Precision SRAM-Based CIM Accelerator With High Macro Utilization Using Structured Pruning Mixed-Precision Networks

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Chen, Yen-Wen; Wang, Rui-Hsuan; Cheng, Yu-Hsiang; Lu, Chih-Cheng; Chang, Meng-Fan; Tang, Kea-Tiong

被引用次数: 1

2024

Past, Present, and Future of CASS Educational Programs and Initiatives [Feature]

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Rokhani, Fakhrol Zaman; Zhang, Xinmiao; Joshi, Rajiv, V; Reis, Ricardo; Grimblatt, Victor; Tang, Kea-Tiong; Li, Yongfu; Amara, Amara; Delgado-Restituto, Manuel

被引用次数: 0

2024

Implementation of a Double-Path Flipped Voltage Follower Cell for Voltage Conveyors

IEEE ACCESS

Akbari, Meysam; Tang, Kea-Tiong

被引用次数: 0

2024

Special Section on the 75th Anniversary of the IEEE Circuits and Systems Society

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Li, Yongfu; Rokhani, Fakhrol Zaman; Lee, Hanho; Tang, Kea-Tiong; Chen, Jie; Ismail, Mohd Hafiz; Uwate, Yoko; Brackin, Molly; Amara, Amara; Sunwoo, Myung Hoon

被引用次数: 0

2024

Gas Identification Algorithm Based on Dynamic Response Analysis of Metal Oxide Sensors Under Temperature Modulation

IEEE ACCESS

Fan, Ya-Han; Chou, Ting-, I; Chiu, Shih-Wen; Tang, Kea-Tiong

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2024

A 0.9 V Adaptive Sampling Rate Differential Level Crossing-SAR ADC for Biomedical Signal Acquisition System

2024 IEEE THE 20TH ASIA PACIFIC CONFERENCE ON CIRCUITS AND SYSTEMS, APCCAS 2024

Lin, Jui-Ting; Liao, Yi-Hsin; Akbari, Meysam; Tang, Kea-Tiong

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2024

A 22 nm Floating-Point ReRAM Compute-in-Memory Macro Using Residue-Shared ADC for AI Edge Device

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hsu, Hung-Hsi; Wen, Tai-Hao; Khwa, Win-San; Huang, Wei-Hsing; Ke, Zhao-En; Chin, Yu-Hsiang; Wen, Hua-Jin; Chang, Yu-Chen; Hsu, Wei-Ting; Lele, Ashwin Sanjay; Zhang, Bo; Wu, Ping-Sheng; Lo, Chung-Chuan; Liu, Ren-Shuo; Hsieh, Chih-Cheng; Tang, Kea-Tiong; Teng, Shih-Hsin; Chou, Chung-Cheng; Chih, Yu-Der; Chang, Tsung-Yung Jonathan; Chang, Meng-Fan

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2024

A 0.67-to-5.4 TSOPs/W Spiking Neural Network Accelerator With 128/256 Reconfigurable Neurons and Asynchronous Fully Connected Synapses

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Qi, Xiang'ao; Li, Xiangting; Lou, Yuqing; Li, Yongfu; Wang, Guoxing; Tang, Kea-Tiong; Zhao, Jian

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2024

ANP-I: A 28-nm 1.5-pJ/SOP Asynchronous Spiking Neural Network Processor Enabling Sub-0.1- μ J/Sample On-Chip Learning for Edge-AI Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Jilin; Huo, Dexuan; Zhang, Jian; Qian, Chunqi; Liu, Qi; Pan, Liyang; Wang, Zhihua; Qiao, Ning; Tang, Kea-Tiong; Chen, Hong

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2024

HAF: A High PE Utilization Spike-Based CNN Accelerator with Adaptive Input Compression and Forward-Based Data Flow

2024 IEEE BIOMEDICAL CIRCUITS AND SYSTEMS CONFERENCE, BIOCAS 2024

Lee, Kuan-Pei; Wu, Zong-Zhe; Tang, Kea-Tiong

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2024

A Dual-Path Deep Learning Model for Low-Cost Air Quality Sensor Calibration

IEEE SENSORS JOURNAL

Liu, Pang-Chun; Chou, Ting-I.; Chiu, Shih-Wen; Tang, Kea-Tiong

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2024

A 69MHz-Bandwidth 40V/ μ s-Slew-Rate3nV/ $\sqrt$ Hz-Noise 4.5 μ V-Offset ChopperOperational Amplifier

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Koolivand, Yarallah; Rezaeiyan, Yasser; Zamani, Milad; Akbari, Meysam; Shoaee, Omid; Tang, Kea-Tiong; Moradi, Farshad

被引用次数: 0

2024

A 0.2-pJ/SOP Digital Spiking Neuromorphic Processor with Temporal Parallel Dataflow and Efficient Synapse Memory Compression

2024 IEEE 6TH INTERNATIONAL CONFERENCE ON AI CIRCUITS AND SYSTEMS, AICAS 2024

Lin, Yu-Hsuan; Tang, Kea-Tiong

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2024

The Conjugated Current Mirrors: A General Enhancement in Transconductance Amplifiers

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Akbari, Meysam; Tang, Kea-Tiong

被引用次数: 0

2024

Cross-site validation of lung cancer diagnosis by electronic nose with deep learning: a multicenter prospective study

RESPIRATORY RESEARCH

Lee, Meng-Rui; Kao, Mu-Hsiang; Hsieh, Ya-Chu; Sun, Min; Tang, Kea-Tiong; Wang, Jann-Yuan; Ho, Chao-Chi; Shih, Jin-Yuan; Yu, Chong-Jen

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2024

8-Bit Precision 6T SRAM Compute-in-Memory Macro Using Global Bitline-Combining Scheme for Edge AI Chips

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Su, Jian-Wei; Lu, Pei-Jung; Wu, Ping-Chun; Chou, Yen-Chi; Liu, Ta-Wei; Chung, Yen-Lin; Hung, Li-Yang; Ren, Jin-Sheng; Huang, Wei-Hsing; Chien, Chih-Han; Mei, Peng-, I; Li, Sih-Han; Sheu, Shyh-Shyuan; Lo, Wei-Chung; Chang, Shih-Chieh; Hong, Hao-Chiao; Lo, Chung-Chuan; Liu, Ren-Shuo; Hsieh, Chih-Cheng; Tang, Kea-Tiong; Chang, Meng-Fan

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2024

A Low-Cost Air Quality Sensor Calibration Algorithm Using Self-Attention Network

2024 IEEE INTERNATIONAL SYMPOSIUM ON OLFACTION AND ELECTRONIC NOSE, ISOEN

Liu, Pang-Chun; Chou, Ting-I; Chiu, Shih-Wen; Tang, Kea-Tiong

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2024

A Low-Noise, Low-Power Neural Signal Amplifier for Deep Brain Stimulation System Chips Tolerating 3V Stimulation

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Hsu, Chia-Hua; Lin, Yu -Wei; Tang, Kea-Tiong

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2024

GEM: A Generalized Memristor Device Modeling Framework Based on Neural Network for Transient Circuit Simulation

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhang, Yuhang; He, Guanghui; Tang, Kea-Tiong; Li, Yongfu; Wang, Guoxing

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2023

Implementation of a Multipath Fully Differential OTA in 0.18- μ m CMOS Process

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Akbari, Meysam; Hussein, Safwan Mawlood; Hashim, Yasir; Khateb, Fabian; Kulej, Tomasz; Tang, Kea-Tiong

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2023

An Aging Drift Calibration and Device-Generality Network With Realistic Transfer Samples for Electronic Nose

IEEE SENSORS JOURNAL

Chou, Ting-, I; Hsueh, Chien-Fu; Yang, Kung-Hsung; Chiu, Shih-Wen; Kuo, Han-Wen; Tang, Kea-Tiong

被引用次数: 1

2023

Development of a Nondestructive Moldy Coffee Beans Detection System Based on Electronic Nose

IEEE SENSORS LETTERS

Tang, Chang-Lin; Chou, Ting-, I; Yang, Sang-Ren; Lin, Yi-Jhen; Ye, Zhong-Kai; Chiu, Shih-Wen; Lee, Sheng-Wei; Tang, Kea-Tiong

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2023

A Rail-to-Rail Transconductance Amplifier Based on Current Generator Circuits

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Akbari, Meysam; Hussein, Safwan Mawlood; Hashim, Yasir; Khateb, Fabian; Tang, Kea-Tiong

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2023

A 0.8 V Intelligent Vision Sensor With Tiny Convolutional Neural Network and Programmable Weights Using Mixed-Mode Processing-in-Sensor Technique for Image Classification

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hsu, Tzu-Hsiang; Chen, Guan-Cheng; Chen, Yi-Ren; Liu, Ren-Shuo; Lo, Chung-Chuan; Tang, Kea-Tiong; Chang, Meng-Fan; Hsieh, Chih-Cheng

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2023

A Multimode Vision Sensor With Temporal Contrast Pixel and Column-Parallel Local Binary Pattern Extraction for Dynamic Depth Sensing Using Stereo Vision

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chiu, Min-Yang; Chen, Guan-Cheng; Hsu, Tzu-Hsiang; Liu, Ren-Shuo; Lo, Chung-Chuan; Tang, Kea-Tiong; Chang, Meng-Fan; Hsieh, Chih-Cheng

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2023

A 5.3 pJ/Spike CMOS Neural Array Employing Time-Modulated Axon-Sharing and Background Mismatch Calibration Techniques

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Qi, Xiang'ao; Zhao, Jian; Lou, Yuqing; Wang, Guoxing; Tang, Kea-Tiong; Li, Yongfu

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2023

Gas Prediction Method Based on Dynamic Response Analysis of Metal Oxide Sensors under Temperature Modulation

2023 IEEE SENSORS

Fan, Ya-Han; Chou, Ting-I; Chiu, Shih-Wen; Tang, Kea-Tiong

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2023

Electronic nose in differentiating and ascertaining clinical status among patients with pulmonary nontuberculous mycobacteria: A prospective multicenter study

JOURNAL OF INFECTION

Lee, Meng-Rui; Huang, Hung-Ling; Huang, Wei-Chang; Wu, Shang-Yu; Liu, Pang-Chun; Wu, Jui-Ching; Cheng, Meng-Hsuan; Sheu, Chau-Chyun; Tang, Kea-Tiong; Wang, Jann-Yuan; Ho, Chao-Chi; Shih, Jin-Yuan; Chong, Inn-Wen

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2023

A 4.5 μ W Miniaturized 3-Channel Wireless Intra-Cardiac Acquisition System

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Rezaeiyan, Yasser; Koolivand, Yarallah; Zamani, Milad; Shoaie, Omid; Akbari, Meysam; Moradi, Farshad; Tang, Kea-Tiong

被引用次数: 0

2023

An Area-Efficient Accelerator for Non-Maximum Suppression

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Sun, Kaihang; Li, Zhaofang; Zheng, Yangfan; Kuo, Hao-Wen; Lee, Kuan-Pei; Tang, Kea-Tiong

被引用次数: 2

2023

Chemiresistive room temperature NO₂ sensor based on nitrogen doped zinc oxide nanowires

SENSORS AND ACTUATORS B-CHEMICAL

Shihabudeen, P. K.; Gupta, Shivam; Notash, Mina Yaghoobi; Sardroodi, Jaber Jahanbin; Chiu, Shih-Wen; Tai, Nyan-Hwa; Tang, Kea-Tiong

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2023

Deep neural network of E-nose sensor for lung cancer classification

2023 IEEE BIOSENSORS CONFERENCE, BIOSENSORS

Kao, Mu-Hsiang; Chiu, Shih-Wen; Lee, Meng-Rui; Sun, Min; Tang, Kea-Tiong

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2023

A 0.5-V Multiple-Input Bulk-Driven OTA in 0.18- μ m CMOS

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Khateb, Fabian; Kulej, Tomasz; Akbari, Meysam; Tang, Kea-Tiong

被引用次数: 22

2022

0.4-V Tail-Less Quasi-Two-Stage OTA Using a Novel Self-Biasing Transconductance Cell

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Akbari, Meysam; Hussein, Safwan Mawlood; Hashim, Yasir; Tang, Kea-Tiong

被引用次数: 13

2022

Low-Voltage Implementation of Neuromorphic Circuits for a Spike-Based Learning Control Module

IEEE ACCESS

Akbari, Meysam; Tang, Kea-Tiong

被引用次数: 0

2022

MARS: Multimacro Architecture SRAM CIM-Based Accelerator With Co-Designed Compressed Neural Networks

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Sie, Syuan-Hao; Lee, Jye-Luen; Chen, Yi-Ren; Yeh, Zuo-Wei; Li, Zhaofang; Lu, Chih-Cheng; Hsieh, Chih-Cheng; Chang, Meng-Fan; Tang, Kea-Tiong

被引用次数: 10

2022

A Transferable Feature-Based Classifier to Improve Transferability of Electronic Nose Systems

IEEE SENSORS LETTERS

Lin, Yi-Jhen; Chou, Ting-, I; Chiu, Shih-Wen; Tang, Kea-Tiong

被引用次数: 3

2022

A 1.93TOPS/W Deep Learning Processor with a Reconfigurable Processing Element array based on SRAM Access Optimization

2022 IEEE ASIA PACIFIC CONFERENCE ON CIRCUITS AND SYSTEMS, APCCAS

Chen, Liao-Chuan; Li, Zhaofang; Lin, Yi-Jhen; Lee, Kuan-Pei; Tang, Kea-Tiong

被引用次数: 0

2022

A Two-stage Training Framework for Hardware Constraints of Computing-in-Memory Architecture

2022 IEEE ASIA PACIFIC CONFERENCE ON CIRCUITS AND SYSTEMS, APCCAS

Kuo, Hao-Wen; Wang, Rui-Hsuan; Li, Zhaofang; Lin, Shih-Ting; Chang, Meng-Fan; Tang, Kea-Tiong

被引用次数: 0

2022

A CMOS Axon-sharing Neuron Array with Background Calibration

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Qi, Xiangao; Zhao, Jian; Wang, Guoxing; Tang, Kea-Tiong; Li, Yongfu

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2022

Two-Way Transpose Multibit 6T SRAM Computing-in-Memory Macro for Inference-Training AI Edge Chips

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Su, Jian-Wei; Si, Xin; Chou, Yen-Chi; Chang, Ting-Wei; Huang, Wei-Hsing; Tu, Yung-Ning; Liu, Ruhui; Lu, Pei-Jung; Liu, Ta-Wei; Wang, Jing-Hong; Chung, Yen-Lin; Ren, Jin-Sheng; Chang, Fu-Chun; Wu, Yuan; Jiang, Hongwu; Huang, Shanshi; Li, Sih-Han; Sheu, Shyh-Shyuan; Wu, Chih-, I; Lo, Chung-Chuan; Liu, Ren-Shuo; Hsieh, Chih-Cheng; Tang, Kea-Tiong; Yu, Shimeng; Chang, Meng-Fan

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2022

A Review of Emerging Electromagnetic-Acoustic Sensing Techniques for Healthcare Monitoring

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Fang, Zhongyuan; Gao, Fei; Jin, Haoran; Liu, Siyu; Wang, Wensong; Zhang, Ruochong; Zheng, Zesheng; Xiao, Xuan; Tang, Kai; Lou, Liheng; Tang, Kea-Tiong; Chen, Jie; Zheng, Yuanjin

被引用次数: 7

2022

Using a Hybrid Deep Neural Network for Gas Classification

IEEE SENSORS JOURNAL

Wang, Syuan-He; Chou, Ting-, I; Chiu, Shih-Wen; Tang, Kea-Tiong

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2021

Low-Hardware-Cost SNN employing FeFET-based Neurons with Tunable Leaky Effect

2021 IEEE BIOMEDICAL CIRCUITS AND SYSTEMS CONFERENCE (IEEE BIOCAS 2021)

Liu, Hongyi; Qi, Xiangao; Lou, Yuqing; Qi, Liang; Yeh, Zuo-Wei; Tang, Kea-Tiong; Zhao, Jian

被引用次数: 0

2021

A Bio-Inspired Motion Detection Circuit Model for the Computation of Optical Flow: The Spatial-Temporal Filtering Reichardt Model

2021 IEEE 3RD INTERNATIONAL CONFERENCE ON ARTIFICIAL INTELLIGENCE CIRCUITS AND SYSTEMS (AICAS)

Wu, Hsin-Yu; Kao, Wei-Tse; Ku, Harrison Hao-Yu; Wang, Cheng-Te; Hsieh, Chih-Cheng; Liu, Ren-Shuo; Tang, Kea-Tiong; Lo, Chung-Chuan

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2021

Integer Quadratic Integrate-and-Fire (IQIF): A Neuron Model for Digital Neuromorphic Systems

2021 IEEE 3RD INTERNATIONAL CONFERENCE ON ARTIFICIAL INTELLIGENCE CIRCUITS AND SYSTEMS (AICAS)

Wu, Wen-Chieh; Yeh, Chen-Fu; White, Alexander James; Wang, Cheng-Te; Yeh, Zuo-Wei; Hsieh, Chih-Cheng; Liu, Ren-Shuo; Tang, Kea-Tiong; Lo, Chung-Chuan

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2021

An Enhanced Input Differential Pair for Low-Voltage Bulk-Driven Amplifiers

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Akbari, Meysam; Hussein, Safwan Mawlood; Hashim, Yasir; Tang, Kea-Tiong

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2021

A 0.5-V Real-Time Computational CMOS Image Sensor With Programmable Kernel for Feature Extraction

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hsu, Tzu-Hsiang; Chen, Yi-Ren; Liu, Ren-Shuo; Lo, Chung-Chuan; Tang, Kea-Tiong; Chang, Meng-Fan; Hsieh, Chih-Cheng

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2021

A 0.8 V Multimode Vision Sensor for Motion and Saliency Detection With Ping-Pong PWM Pixel

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hsu, Tzu-Hsiang; Chen, Yen-Kai; Chiu, Min-Yang; Chen, Guan-Cheng; Liu, Ren-Shuo; Lo, Chung-Chuan; Tang, Kea-Tiong; Chang, Meng-Fan; Hsieh, Chih-Cheng

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A 0.3-V Conductance-Based Silicon Neuron in 0.18 μm CMOS Process

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Akbari, Meysam; Hussein, Safwan Mawlood; Chou, Ting-, I; Tang, Kea-Tiong

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2021

An Adjustable Dual-Output Current Mode MOSFET-Only Filter

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Akbari, Meysam; Hussein, Safwan Mawlood; Hashim, Yasir; Tang, Kea-Tiong

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2021

A 4-Kb 1-to-8-bit Configurable 6T SRAM-Based Computation-in-Memory Unit-Macro for CNN-Based AI Edge Processors

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chiu, Yen-Cheng; Zhang, Zhixiao; Chen, Jia-Jing; Si, Xin; Liu, Ruhui; Tu, Yung-Ning; Su, Jian-Wei; Huang, Wei-Hsing; Wang, Jing-Hong; Wei, Wei-Chen; Hung, Je-Min; Sheu, Shyh-Shyuan; Li, Sih-Han; Wu, Chih-I; Liu, Ren-Shuo; Hsieh, Chih-Cheng; Tang, Kea-Tiong; Chang, Meng-Fan

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2020

A Twin-8T SRAM Computation-in-Memory Unit-Macro for Multibit CNN-Based AI Edge Processors

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Si, Xin; Chen, Jia-Jing; Tu, Yung-Ning; Huang, Wei-Hsing; Wang, Jing-Hong; Chiu, Yen-Cheng; Wei, Wei-Chen; Wu, Ssu-Yen; Sun, Xiaoyu; Liu, Rui; Yu, Shimeng; Liu, Ren-Shuo; Hsieh, Chih-Cheng; Tang, Kea-Tiong; Li, Qiang; Chang, Meng-Fan

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2020

Editorial

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Wang, Guoxing; Constandinou, Timothy G.; Tang, Kea-Tiong

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2020

A Minimum Distance Inlier Probability (MDIP) Feature Selection Method to Improve Gas Classification for Electronic Nose Systems

IEEE ACCESS

Liu, Yen-Tung; Tang, Kea-Tiong

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2020

A Fully Integrated High-Power-Supply-Rejection Linear Regulator With an Output-Supplied Voltage Reference

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chen, Yan-Peng; Tang, Kea-Tiong

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2020

Embedded 1-Mb ReRAM-Based Computing-in-Memory Macro With Multibit Input and Weight for CNN-Based AI Edge Processors

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Xue, Cheng-Xin; Chen, Wei-Hao; Liu, Je-Syu; Li, Jia-Fang; Lin, Wei-Yu; Lin, Wei-En; Wang, Jing-Hong; Wei, Wei-Chen; Huang, Tsung-Yuan; Chang, Ting-Wei; Chang, Tung-Cheng; Kao, Hui-Yao; Chiu, Yen-Cheng; Lee, Chun-Ying; King, Ya-Chin; Lin, Chrong-Jung; Liu, Ren-Shuo; Hsieh, Chih-Cheng; Tang, Kea-Tiong; Chang, Meng-Fan

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2020

An Energy-Efficient SAR ADC With Event-Triggered Error Correction

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ou-Yang, Yi-Han; Tang, Kea-Tiong

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2019

Detection of Cigarette Smoke Using a Surface-Acoustic-Wave Gas Sensor with Non-Polymer-Based Oxidized Hollow Mesoporous Carbon Nanospheres

MICROMACHINES

Cheng, Chi-Yung; Huang, Shih-Shien; Yang, Chia-Min; Tang, Kea-Tiong; Yao, Da-Jeng

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2019

CMOS-integrated memristive non-volatile computing-in-memory for AI edge processors

NATURE ELECTRONICS

Chen, Wei-Hao; Dou, Chunmeng; Li, Kai-Xiang; Lin, Wei-Yu; Li, Pin-Yi; Huang, Jian-Hao; Wang, Jing-Hong; Wei, Wei-Chen; Xue, Cheng-Xin; Chiu, Yen-Cheng; King, Ya-Chin; Lin, Chrong-Jung; Liu, Ren-Shuo; Hsieh, Chih-Cheng; Tang, Kea-Tiong; Yang, J. Joshua; Ho, Mon-Shu; Chang, Meng-Fan

A Gas Mixture Prediction Model Based on the Dynamic Response of a Metal-Oxide Sensor MICROMACHINES

Wen, Wei-Chih; Chou, Ting-I; Tang, Kea-Tiong

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2019

A 1-V 2.6-mW Environmental Compensated Fully Integrated Nose-on-a-Chip

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Chou, Ting-I; Chang, Kwang-Han; Jhang, Jia-Yin; Chiu, Shih-Wen; Wang, Guoxing; Yang, Chia-Hsiang; Chiu, Herming; Chen, Hsin; Hsieh, Chih-Cheng; Chang, Meng-Fan; Tang, Kea-Tiong

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2018

Development of a Dual MOS Electronic Nose/Camera System for Improving Fruit Ripeness Classification

SENSORS

Chen, Li-Ying; Wu, Cheng-Chun; Chou, Ting-I; Chiu, Shih-Wen; Tang, Kea-Tiong

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2018

Highly Sensitive and Portable Gas Sensing System Based on Reduced Graphene Oxide

TSINGHUA SCIENCE AND TECHNOLOGY

Mi, Wentian; Chiu, Shih-Wen; Xue, Tao; Chen, Yuanquan; Qi, Hanyu; Yang, Yi; Tang, Kea-Tiong; Ren, Tian-Ling

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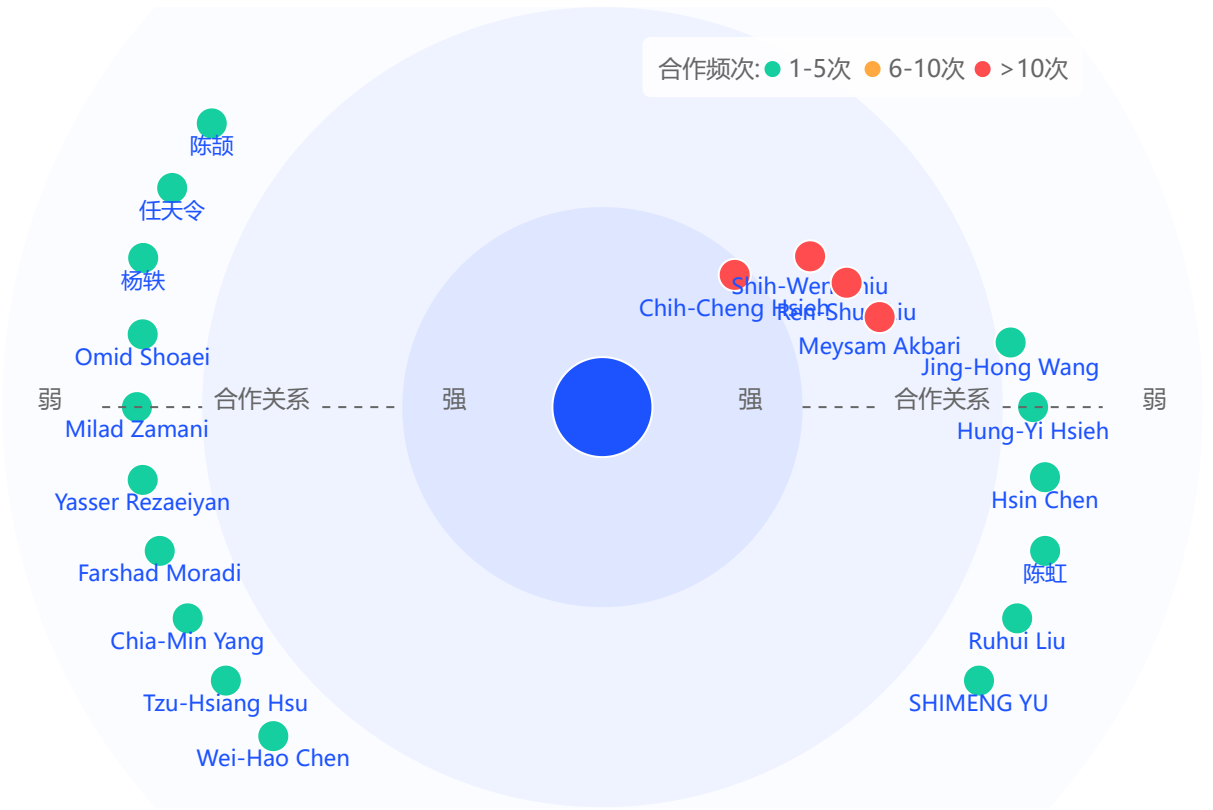
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